A Sharp Perturbation Dichotomy for Independent-Island Phase Retrieval

Run `fa_banach_001`, lane 11

22 July 2026

Status. Candidate sharp dichotomy (positive theorem plus counterexample), likely valid, for human review.

1 Source question

Alharbi, Freeman, Ghoreishi, Lois, and Sebastian prove that ordinary stable phase retrieval in a finite-dimensional Hilbert space persists under a small perturbation of the frame vectors. They then discuss signals supported on separated islands, where recovery only up to an independent phase on each island is sufficient, and state that the effect of perturbations in this setting remains an important open problem [1, Introduction, PDF p. 3].

> to a subspace of ℓ_4^n [FLM], and thus for all $a_0 > 0$ and $k \in \mathbb{N}$ there exists $n \in \mathbb{N}$ so that every Parseval frame $(x_j)_{j=1}^{kn}$ of ℓ_2^n fails (1.3) for the value a_0. On the other hand, working in $\ell_4(J)$ or more generally $L_4(\mu)$ allows for the introduction of some powerful analytic methods [B][CPT].
>
> Theorem 1.2 gives a solution to the problem of determining how perturbation affects the stability of phase retrieval of frames for finite dimensional Hilbert spaces. However, phase retrieval is often considered in more general settings, and there are many opportunities for considering the effect of perturbations on phase retrieval. In applications one is usually interested in objects with some particular structure and it is not necessary that (1.2) be satisfied for all $x, y \in H$, but only for x and y in some subset of interest. Furthermore, it is often permissible to consider a weaker equivalence relation than $x \sim y$ if and only if $x = \lambda y$ for some $|\lambda| = 1$. For example, consider f and g to be audio recordings where the audio for f stops a full second before the audio for g starts. Then $f+g$, $f-g$, $-f+g$, and $-f-g$ would all sound the same and would all have the same absolute value. Thus, we may consider larger equivalence classes as we would be satisfied with obtaining any of those vectors when doing phase retrieval. This situation arises in many important contexts when studying phase retrieval and it is often possible to stably reconstruct the components of a signal which are supported on separated islands (even though it is not possible to determine the relative phase between different components) [ADGY][CCSW][CDDL][FKM][GR][GR2]. The problem of how stability is affected by perturbations in these kinds of circumstances remains an important open problem. It is known that if a frame does norm retrieval and not phase retrieval then norm retrieval is not stable under perturbations [HR], and it would be interesting to know how perturbations affect phase retrieval by projections [CGJT][EGK] and weak phase retrieval [BCGJT] as well. In [CDFF][CPT][FOPT], stable phase retrieval is studied for infinite dimensional subspaces of Banach lattices, and the problem of how stability of phase retrieval is affected by perturbations is considered in Section 4 of [FOPT]. There are many opportunities for further research on how this may be quantified and how different Banach space geometry allows for more refined perturbation estimates. We hope that this paper provides some inspiration to further study the relationship between perturbation and the stability of phase retrieval.

The question is a research program rather than a single quantified conjecture. We formalize its basic orthogonal-island model and obtain a sharp dichotomy. Stable recovery need not persist under arbitrary perturbations, but it does persist—with the same explicit dimension-free estimate as in the source theorem—when every measurement vector remains in its original island.

2 Claimed contribution

Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$, and put $H = \mathbb{F}^2$ with the orthogonal island decomposition

$$H = H_1 \oplus H_2, \qquad H_1 = \mathbb{F}e_1, \quad H_2 = \mathbb{F}e_2.$$

Define $x \sim_{\mathrm{isl}} y$ when there are $\lambda_1, \lambda_2 \in \mathbb{T}$ (with $\mathbb{T} = \{-1, 1\}$ over $\mathbb{R}$) such that

$$x_k = \lambda_k y_k \qquad (k = 1, 2).$$

Thus relative phase between the two islands is deliberately ignored. Equip the quotient with

$$d_{\mathrm{isl}}(x, y) = \min_{\lambda_1, \lambda_2 \in \mathbb{T}} \left(|x_1 - \lambda_1 y_1|^2 + |x_2 - \lambda_2 y_2|^2\right)^{1/2}.$$

Theorem 1. *The frame $X = (e_1, e_2)$ does 1-stable phase retrieval modulo $\sim_{\mathrm{isl}}$. For every $\eta > 0$, however, there is a frame $Y = (y_1, y_2)$ satisfying*

$$\sum_{j=1}^{2} \|y_j - e_j\|^2 < \eta$$

which is not even injective modulo $\sim_{\mathrm{isl}}$. Consequently no analogue of arbitrary-perturbation stability is possible for a fixed independent-island quotient, even in dimension two.

3 Proof intuition

The coordinate frame measures the two islands separately, so its magnitude map is exactly the quotient-coordinate map $(x_1, x_2) \mapsto (|x_1|, |x_2|)$. Mix a tiny amount of the second island into the first measurement. Flipping the sign of that mixed measurement and solving the resulting two linear equations produces another signal with identical magnitudes. The new signal has a nonzero first-island component, whereas the original one does not, so the two signals are not equivalent under independent island phases. The instability is therefore qualitative, not merely a loss in the Lipschitz constant.

Conversely, if measurement vectors are forbidden from crossing between islands, the problem separates exactly. Restricting the original stability inequality to one island gives ordinary stable phase retrieval there. The source perturbation theorem applies on every island, and orthogonality lets us sum the squared estimates without losing either the threshold or the stability constant.

4 Counterexample proof

For scalars a, b,

$$\min_{\lambda \in \mathbb{T}} |a - \lambda b| = \big||a| - |b|\big|.$$

It follows coordinatewise that

$$d_{\mathrm{isl}}(x, y)^2 = \big(|x_1| - |y_1|\big)^2 + \big(|x_2| - |y_2|\big)^2. \tag{1}$$

The analysis map of $X = (e_1, e_2)$ is $\Theta_X x = (x_1, x_2)$, hence

$$d_{\mathrm{isl}}(x, y) = \big\| |\Theta_X x| - |\Theta_X y| \big\|_2.$$

This proves exact 1-stability modulo $\sim_{\mathrm{isl}}$.

Fix $\eta > 0$ and choose $0 < \varepsilon < \sqrt{\eta}$. Set

$$y_1 = e_1 + \varepsilon e_2, \qquad y_2 = e_2.$$

Then

$$\|y_1 - e_1\|^2 + \|y_2 - e_2\|^2 = \varepsilon^2 < \eta.$$

The analysis matrix of $Y = (y_1, y_2)$ is

$$S_\varepsilon = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 \end{pmatrix},$$

so Y remains a basis (and therefore a frame) for every ε.

Now take

$$x = e_2 = (0, 1), \qquad z = -2\varepsilon e_1 + e_2 = (-2\varepsilon, 1).$$

Both vectors are real, so the same calculation works over either scalar field. Directly,

$$S_\varepsilon x = (\varepsilon, 1), \qquad S_\varepsilon z = (-\varepsilon, 1).$$

Thus

$$|\Theta_Y x| = |\Theta_Y z| = (\varepsilon, 1).$$

On the other hand, equation (1) gives

$$d_{\mathrm{isl}}(x, z) = \big((0 - 2\varepsilon)^2 + (1 - 1)^2\big)^{1/2} = 2\varepsilon > 0.$$

Therefore $x \not\sim_{\mathrm{isl}} z$ despite identical magnitude measurements. The perturbed frame fails injectivity on the quotient, so it cannot satisfy any finite stability inequality. This proves the theorem.

5 Structure-preserving perturbations

We now prove the positive half in the natural finite-dimensional orthogonal-island model. Let

$$H = \bigoplus_{k=1}^{r} H_k$$

be an orthogonal decomposition, let P_k denote the orthogonal projection onto H_k, and partition the frame index set as $J = J_1 \sqcup \cdots \sqcup J_r$. Define the independent-island quotient metric by

$$d_r(x, y) = \min_{\lambda_1, \ldots, \lambda_r \in \mathbb{T}} \left\| x - \sum_{k=1}^{r} \lambda_k P_k y \right\|. \tag{2}$$

Theorem 2 (Island-preserving perturbation theorem). *Let $X = (x_j)_{j\in J}$ be a frame for H with frame bounds $B \geq A > 0$. Assume that*

$$x_j \in H_k \quad (j \in J_k)$$

and that X does C-stable phase retrieval for the quotient metric d_r:

$$d_r(x, y) \leq C \left(\sum_{j\in J} \big| |\langle x, x_j\rangle| - |\langle y, x_j\rangle| \big|^2 \right)^{1/2}.$$

Let

$$0 < \nu < \frac{1}{16C^4B},$$

and let $Y = (y_j)_{j\in J}$ be a structure-preserving perturbation satisfying

$$y_j \in H_k \quad (j \in J_k), \qquad \sum_{j\in J} \|x_j - y_j\|^2 < \nu.$$

Then Y is a frame with bounds

$$A\left(1 - \sqrt{\frac{\nu}{A}}\right)^2 \quad \text{and} \quad B\left(1 + \sqrt{\frac{\nu}{B}}\right)^2,$$

and it does C_ν-stable phase retrieval modulo independent island phases, where

$$C_\nu = C\left(1 - 4C^2\sqrt{\nu B}\right)^{-1/2}.$$

Thus the threshold and stability estimate are exactly those of the ordinary perturbation theorem in the source.

Proof. For $u, v \in H_k$, equation (2) reduces to the ordinary phase quotient:

$$d_r(u, v) = \min_{\lambda\in\mathbb{T}} \|u - \lambda v\|.$$

Measurements indexed outside J_k vanish on H_k. Hence the restriction $X_k = (x_j)_{j\in J_k}$ is a frame for H_k with the same admissible frame bounds A, B, and the assumed quotient inequality shows that X_k does C-stable ordinary phase retrieval.

Put $Y_k = (y_j)_{j\in J_k}$. Its perturbation size satisfies

$$\sum_{j\in J_k} \|x_j - y_j\|^2 < \nu.$$

The ordinary perturbation theorem of Alharbi et al. therefore applies to every pair X_k, Y_k with the same parameter ν. It gives the displayed frame bounds and, for all $u, v \in H_k$,

$$\min_{\lambda\in\mathbb{T}} \|u - \lambda v\| \leq C_\nu \left(\sum_{j\in J_k} \big| |\langle u, y_j\rangle| - |\langle v, y_j\rangle| \big|^2 \right)^{1/2}. \tag{3}$$

Summing the block frame inequalities proves the asserted global frame bounds for Y. Finally, orthogonality makes the minimization in (2) separate:

$$d_r(x,y)^2 = \sum_{k=1}^{r} \min_{\lambda\in\mathbb{T}} \|P_k x - \lambda P_k y\|^2.$$

Apply the square of (3) with $u = P_k x$, $v = P_k y$, and sum over k. Since $y_j \in H_k$ for $j \in J_k$, the right-hand side becomes

$$C_\nu^2 \sum_{j\in J} \big| |\langle x, y_j\rangle| - |\langle y, y_j\rangle| \big|^2.$$

Taking square roots gives the claimed global stability estimate. □

6 Sharp dichotomy and scope

The example embeds in $\mathbb{F}^m$ for every $m \geq 2$: retain the same first two frame vectors and signals, and append $e_3, \ldots, e_m$. Hence neither dimension nor the number of separated islands repairs the obstruction.

Theorem 1 and Theorem 2 give the promised sharp answer in the orthogonal-island model. Allowing even one arbitrarily small cross-island component can destroy quotient injectivity, whereas every sufficiently small island-preserving perturbation retains stable recovery with an explicit dimension-free constant. More general notions of approximately separated islands, signal-dependent island decompositions, or perturbations that change the quotient itself remain separate questions.

7 Exact verification

The script `code/verify_counterexample.py` checks with exact rational arithmetic that the perturbation size is nonzero and arbitrarily scalable, the perturbed vectors form a basis, the two displayed magnitude measurements agree, and their quotient distance is positive. The positive theorem is an exact blockwise consequence of the source perturbation theorem; its only additional identity is the orthogonal sum-of-squares decomposition displayed in the proof.

References

[1] W. Alharbi, D. Freeman, D. Ghoreishi, C. Lois, and S. Sebastian, *Stable phase retrieval and perturbations of frames*, Proc. Amer. Math. Soc. Ser. B 10 (2023), 353–368; arXiv:2212.13681.